

Problem

The current in an RLC circuit is described by

$$\frac{d^2 i}{dt^2} + 10 \frac{di}{dt} + 25i = 0$$

If $i(0) = 10$ A and $di(0)/dt = 0$, find $i(t)$ for $t > 0$.

Step-by-step solution

Step 1 of 1

Given equation

$$\frac{d^2 i}{dt^2} + 10 \frac{di}{dt} + 25i = 0$$

and $i(0) = 10$ and $\frac{di(0)}{dt} = 0$,

comparing the equation with the general equation:

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = 0,$$

$$\frac{R}{L} = 10$$

$$\frac{1}{LC} = 25$$

therefore

$$\alpha = \frac{R}{2L}$$

$$\alpha = \frac{10}{2}$$

$$\alpha = 5$$

$$\omega_0 = \sqrt{\frac{1}{LC}}$$

$$\omega_0 = \sqrt{25}$$

$$\omega_0 = 5$$

Since $\alpha = \omega_0$, the circuit is critically damped.

Therefore the solution will be of the form,

$$i(t) = (A_2 + A_1 t) e^{-\alpha t}$$

now to find out A1 and A2,

at $t = 0$

$$i(0) = A_2$$

Hence $A_2 = 10A$

to find A1 , we differentiate i (t) with respect to 't' and substitute at 't = 0'

$$\frac{di(t)}{dt} = -\alpha(A_2 + A_1 t) e^{-\alpha t} + A_1 e^{-\alpha t}$$

therefore at $t = 0$

$$\frac{di(0)}{dt} = -\alpha A_2 + A_1$$

hence

$$A_1 = \alpha A_2$$

$$A_1 = (5)(10)$$

$$A_1 = 50$$

So, the final solution is:

$$i(t) = (10 + 50t) e^{-5t} A \quad \text{for } t > 0$$